AERIS Expert Review Packet

DRAFT

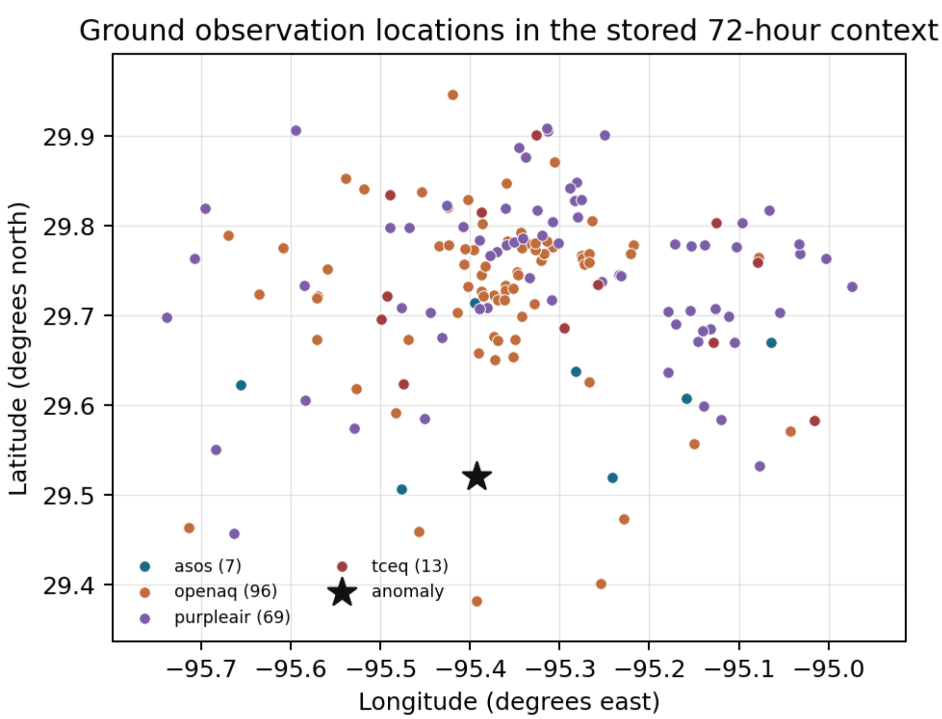
Anomaly

Source / metric	openaq / ozone
Observed / expected	0.04 / 0.0147256
Time	2026-07-08 03:00 UTC
Location	29.5203, -95.3925
Severity	moderate
Detectors	stl, isolation_forest

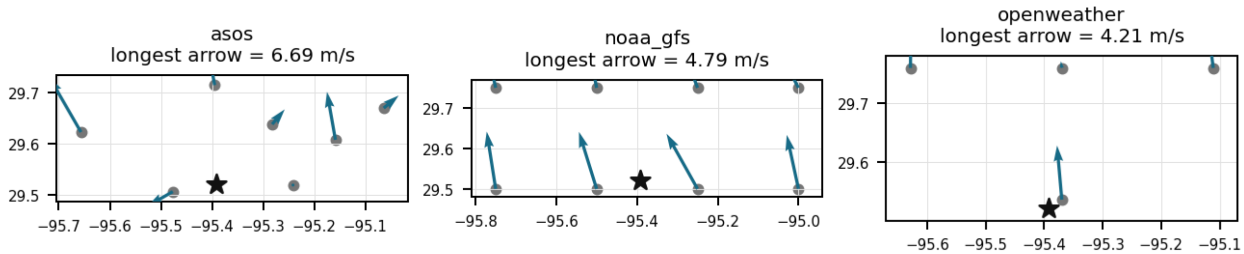
Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	7 / 704	39.12 to 100; mean 73.2391	64.96 at 2026-07-08 02:55Z; dt 5 min
asos	temperature	degC	7 / 704	23 to 36.1111; mean 28.9822	26.1111 at 2026-07-08 02:55Z; dt 5 min
asos	wind_direction	degree	7 / 854	0 to 340; mean 136.897	60 at 2026-07-08 02:55Z; dt 5 min
asos	wind_speed	m/s	7 / 889	0 to 10.8033; mean 2.3529	3.08666 at 2026-07-08 02:55Z; dt 5 min
noaa_gfs	gh_500	m	8 / 96	5896.24 to 5924.75; mean 5910.78	5911.85 at 2026-07-08 00:00Z; dt 180 min
noaa_gfs	pbl_height	m	8 / 96	19.6394 to 2153.73; mean 722.402	1003.53 at 2026-07-08 00:00Z; dt 180 min
noaa_gfs	precipitable_water	mm	8 / 96	39.1305 to 51.5499; mean 46.114	49.6923 at 2026-07-08 00:00Z; dt 180 min
noaa_gfs	surface_pressure	hPa	8 / 96	1008.95 to 1017.68; mean 1014.36	1012.9 at 2026-07-08 00:00Z; dt 180 min
noaa_gfs	t_850	degC	8 / 96	17.4065 to 20.4926; mean 19.0548	18.8821 at 2026-07-08 00:00Z; dt 180 min
noaa_gfs	u_10m	m/s	8 / 96	-2.34889 to 1.76546; mean 0.2332	-1.33889 at 2026-07-08 00:00Z; dt 180 min
noaa_gfs	v_10m	m/s	8 / 96	-0.854414 to 4.60966; mean 2.2122	4.3194 at 2026-07-08 00:00Z; dt 180 min
openaq	ozone	ppm	16 / 726	-0.004 to 0.058; mean 0.017	0.04 at 2026-07-08 03:00Z; dt 0 min
openaq	pm10	ug/m3	3 / 145	8 to 75; mean 25.5793	8 at 2026-07-08 03:00Z; dt 0 min
openaq	pm25	ug/m3	77 / 2668	0 to 62.1; mean 5.3824	4.8 at 2026-07-08 03:00Z; dt 0 min
openweather	cloud_cover	percent	4 / 287	0 to 100; mean 23.0209	24 at 2026-07-08 03:02Z; dt 2 min
openweather	humidity	percent	4 / 287	38 to 95; mean 76.2056	84 at 2026-07-08 03:02Z; dt 2 min
openweather	precipitation	mm/h	4 / 287	0 to 5.62; mean 0.0572	0 at 2026-07-08 03:02Z; dt 2 min
openweather	pressure	hPa	4 / 287	1012 to 1019; mean 1016.36	1017 at 2026-07-08 03:02Z; dt 2 min
openweather	temperature	degC	4 / 287	23.78 to 35.9; mean 29.5798	26 at 2026-07-08 03:02Z; dt 2 min
openweather	wind_direction	degree	4 / 287	0 to 359; mean 193.801	173 at 2026-07-08 03:02Z; dt 2 min
openweather	wind_speed	m/s	4 / 287	0.36 to 8.39; mean 2.2718	3.23 at 2026-07-08 03:02Z; dt 2 min
purpleair	pm25	ug/m3	69 / 4787	0 to 146.4; mean 7.2753	5.9 at 2026-07-08 03:00Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 382.4 min
sentinel5p	s5p_ch4_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 382.4 min
sentinel5p	s5p_co_column	mol/m^2	6 / 6	0.0243723 to 0.0253077; mean 0.0249	0.0249873 at 2026-07-07 18:57Z; dt 482.4 min
sentinel5p	s5p_co_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 382.4 min
sentinel5p	s5p_hcho_column	mol/m^2	8 / 8	0.000174097 to 0.000235341; mean 0.0002	0.000202795 at 2026-07-07 20:37Z; dt 382.4 min
sentinel5p	s5p_hcho_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 382.4 min
sentinel5p	s5p_no2_column	mol/m^2	4 / 4	3.63106e-05 to 4.30727e-05; mean 0	4.10314e-05 at 2026-07-07 20:37Z; dt 382.4 min
sentinel5p	s5p_no2_granule_available	count	4 / 4	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 382.4 min
sentinel5p	s5p_o3_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 382.4 min
sentinel5p	s5p_so2_column	mol/m^2	8 / 8	-3.49304e-05 to 4.72775e-05; mean 0	2.51087e-05 at 2026-07-07 20:37Z; dt 382.4 min
sentinel5p	s5p_so2_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 382.4 min
tceq	co	ppm	3 / 162	0 to 1; mean 0.2383	0.3 at 2026-07-08 03:00Z; dt 0 min
tceq	no2	ppb	12 / 633	-0.1 to 35.8; mean 6.4193	2.9 at 2026-07-08 03:00Z; dt 0 min
tceq	so2	ppb	3 / 154	-0.5 to 6.8; mean -0.0513	-0.4 at 2026-07-08 02:00Z; dt 60 min



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 36

Winds shifted to a northeasterly direction (60 degrees) at speeds of approximately 3 m/s, advecting ozone from upwind industrial areas.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 2 of 36

A rapid collapse of the planetary boundary layer from over 930 meters to approximately 222 meters decoupled the surface layer, while light winds of around 3 m/s allowed daytime ozone to persist into the evening with delayed titration.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 3 of 36

Meteorology favored persistence and transport of an earlier-produced ozone plume: surface winds near the anomaly were light to moderate (~3.1 m/s) with easterly to southeasterly character nearby, and model 10 m flow around 00Z also supported onshore/southerly transport into Houston.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 4 of 36

The anomalous pollutant is ozone at 29.520, -95.392, measured at 0.04 ppm on 2026-07-08T03:00:00, compared with an expected value of 0.014725584278787614 ppm; this was flagged as moderate severity by the detection methods.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 5 of 36

The nearest air quality monitoring station recorded a local nitrogen dioxide (NO_2) concentration of 2.9 ppb concurrent with the ozone anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 6 of 36

High daytime temperatures exceeding 32-34°C and significant formaldehyde and nitrogen dioxide columns on July 7 established a strong photochemical foundation for high ozone levels prior to the anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 7 of 36

High levels of nitrogen dioxide (NO_2) and ozone (O_3) are contributing to the poor air quality.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 8 of 36

The SO_2 levels are not significantly different from normal, ruling out SO_2 as a primary cause.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 9 of 36

The anomaly was detected in the vicinity of Houston, Texas, with a nearest station located approximately 26.788 km away from the anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 10 of 36

Low local nitrogen dioxide levels of 2.9 ppb reduced nighttime titration, allowing the advected ozone to persist.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 11 of 36

Elevated levels of nitrogen dioxide and formaldehyde observed earlier in the day provided the necessary chemical precursors to sustain high daytime ozone production, which then persisted into the late evening.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 12 of 36

The presence of volatile organic compounds (VOCs) is also a possible cause, as they can react with NOx and sunlight to form ground-level ozone.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 13 of 36

Local chemistry and weak removal likely helped sustain the ozone enhancement: there was little precipitation at the event time, low PM2.5 and PM10, and no strong contemporaneous indication of fresh NO titration at the nearest regulatory NO2 site (NO2 near 2.9 ppb at the anomaly time), which is consistent with relatively aged air where ozone can remain elevated into the evening rather than being rapidly destroyed near fresh combustion sources.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 14 of 36

Weather patterns may have also played a role in the air quality anomaly in Houston, Texas.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 15 of 36

The rapid collapse of the planetary boundary layer from over 900 meters to approximately 220 meters, combined with surface wind speeds of around 3 m/s, suggests that downward vertical mixing from the ozone-rich residual layer may have transported ozone to the surface.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 16 of 36

The air quality anomaly in Houston, Texas was characterized by elevated levels of NO2, O3, and SO2.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 17 of 36

A carryover residual-layer ozone plume from earlier daytime photochemistry is a leading hypothesis: area ozone was already elevated in the 00Z–06Z window on 2026-07-08 (6 h mean about 0.025 ppm before dropping later), and the anomaly time had warm evening temperatures with a still moderately deep boundary layer near 00Z that would allow ozone formed earlier to remain aloft and mix to the surface as conditions evolved overnight.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 18 of 36

The anomaly occurred on July 7, 2026, at approximately 20:37 UTC.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 19 of 36

High levels of CO and NO2 are contributing to the air quality anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 20 of 36

Local emissions from industrial activities and vehicle traffic likely contributed to the air quality anomaly in Houston, Texas.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 21 of 36

A moderate nighttime ozone anomaly of 0.04 ppm was detected in Houston at 03:00 UTC on July 8, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 22 of 36

Northeasterly surface winds observed near the anomaly site may have horizontally advected ozone or its precursors from the highly industrial Houston Ship Channel region.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 23 of 36

Weather patterns such as heatwaves or stagnant air masses may be exacerbating the issue by trapping pollutants near the surface.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 24 of 36

There is little evidence at the anomaly time for a fresh combustion, dust, or sulfur-dominated plume driving the signal, because PM2.5 stayed low, satellite CO was not elevated, and nearby SO2 observations were near zero or negative.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 25 of 36

Residual daytime ozone persisting into the evening is strongly supported: the local ozone at 03:00 UTC was 0.04 ppm versus an expected 0.0147 ppm, and the regional 6 h mean ozone for 00Z on 2026-07-08 was already elevated at 0.0252 ppm before collapsing by 06Z to 0.0040 ppm, indicating the anomaly occurred during the evening decay phase rather than as an isolated spike. Warm conditions near 26 C, only partial cloud cover around 24–30%, no rain at the anomaly time, and a still moderately deep evening boundary layer near 933–1004 m are all compatible with delayed loss and downward mixing of ozone produced earlier in the day.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 26 of 36

Purely local fresh precursor accumulation is less favored, though not impossible: contemporaneous nearby NO_2 was low at about 2.9 ppb and SO_2 near zero or negative, TROPOMI columns do not show a strong coincident enhancement in CO, HCHO, or SO_2 before the event, and PM2.5/PM10 were low to modest, arguing against a fresh combustion or sulfur plume driving the ozone. However, overnight humidity was high and winds later weakened markedly, so some local trapping of already aged oxidants or VOC-rich air cannot be excluded.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 27 of 36

The anomaly is related to elevated levels of hydrogen chloride (HCl) and sulfur dioxide (SO_2).

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 28 of 36

At the time of the anomaly, the nearest meteorological station recorded a temperature of 26.11 degC and light winds of 3.09 m/s from an east-northeasterly direction of 60 degrees.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 29 of 36

At the anomaly time, ozone at the site was substantially elevated (0.04 ppm versus expected 0.0147 ppm), making this a real moderate positive evening ozone excursion rather than a small fluctuation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 30 of 36

Within the local monitoring context, openaq ozone 6-hour mean for 2026-07-08 00Z was 0.0252 ppm, indicating the anomaly occurred during an elevated ozone period relative to the broader multi-day mean of 0.017 ppm across nearby ozone stations.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 31 of 36

Transport from urban/industrial precursor sources is also plausible. Near the anomaly, surface winds were modest and from the E/NE at the closest ASOS site (~60 degrees, ~3.1 m/s), while regional modeled 10 m flow around 00Z was toward the north from a southerly component, indicating directional variability consistent with plume meander/recirculation; Houston ozone commonly spikes when aged urban-industrial air is advected rather than being fully dispersed.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 32 of 36

A shift in wind direction toward the east-northeast near the anomaly suggests that horizontal advection of ozone-rich air or precursors from the industrial Ship Channel sector contributed to the elevated concentrations.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 33 of 36

An ozone anomaly of 0.04 ppm was recorded at coordinates (29.520, -95.392) on July 8, 2026, at 03:00:00 UTC, exceeding the expected baseline of approximately 0.015 ppm.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 34 of 36

The air quality anomaly is likely caused by a combination of factors.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 35 of 36

Transport from the Houston urban/industrial corridor is also well supported: near the anomaly, surface winds were light to modest but directional fields show a shift consistent with advection, with ASOS nearest-station winds near 60 degrees at 02:55 UTC, ASOS 00Z mean direction about 117 degrees, and gfs 10 m flow at 00Z indicating southeasterly onshore transport. Such flow can carry aged ozone or precursor-rich air from the Ship Channel/urban core toward the anomaly area while wind speeds around 2.8–3.2 m/s are sufficient for plume movement without rapid dilution.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 36 of 36

At nearly the same time, nearby surface observations showed about 26.1 C temperature, about 65% relative humidity, wind from about 60 degrees at about 3.09 m/s, while nearby PM2.5 was low at about 4.8 to 5.9 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.